Formulas

A formula is a type of equation that shows the relationship between different variables. A variable is a symbol such as "x" or "y" that stands in for an unknown number. Physicists calculate unknown quantities using formulas, in which known quantities are combined in specific ways. Listed below are some of the most common formulas.

Physics formulas

Quantity	Description	Formula
Current	$\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
Voltage	current × resistance	$V = IR$
Resistance	$\frac{\text{voltage}}{\text{current}}$	$R = \frac{V}{I}$
Power	$\frac{\text{work}}{\text{time}}$	$P = \frac{W}{t}$
Time	$\frac{\text{distance}}{\text{velocity}}$	$t = \frac{d}{v}$
Distance	velocity × time	$d = vt$
Velocity	$\frac{\text{displacement (distance in a given direction)}}{\text{time}}$	$v = \frac{d}{t}$
Acceleration	$\frac{\text{final velocity} - \text{initial velocity}}{\text{time}}$	$a = \frac{v2 - v1}{t}$
Force	mass × acceleration	$F = ma$
Momentum	mass × velocity	$p = mv$
Pressure	$\frac{\text{force}}{\text{area}}$	$P = \frac{F}{A}$
Density	$\frac{\text{mass}}{\text{volume}}$	$p = \frac{m}{V}$
Volume	$\frac{\text{mass}}{\text{density}}$	$V = \frac{m}{p}$
Mass	volume × density	$m = Vp$
Area	length × width	$A = lw$
Kinetic energy	½ mass × square of velocity	$E_k = \frac{1}{2} mv^2$
Weight	mass × acceleration due to gravity	$W = mg$
Work done	force × distance moved in direction of force	$W = Fs$

Types of energy

Energy is the ability to make things happen, whether it is moving something, heating it up, or changing it in some way. Energy exists in many forms, including sound, heat, and light. All types of energy are related, and can be converted from one form to another.

Thermal energy
The energy that Earth receives from the Sun is called thermal or heat energy. Air blowing out from a hairdryer is hot because electrical energy is converted into thermal energy.

Chemical energy
This is the form of energy released when a chemical reaction, such as burning fuel, takes place. When food is digested, chemical compounds are broken down and energy is released into the body.

Nuclear energy
Nuclear energy is the potential energy stored in the nucleus of an atom. When the nucleus of an atom is split or when two nuclei fuse together, a tremendous amount of energy is released.

Potential energy
Potential energy is energy that is stored, ready for use. For example a diver has potential energy due to her or his height above the water. This changes to kinetic energy as the diver falls.

Radiant energy
This is the form of energy carried by light and other types of electromagnetic radiation. The Sun is Earth's major source of radiant energy because it gives off vast amounts of heat and light.

Sound energy
Sound energy is produced when an object vibrates. The sound vibrations cause waves that travel through a medium such as air, water, wood, or metal.

Kinetic energy
This is the energy of motion. All moving objects—such as burning fuel atoms to aircraft—possess kinetic energy. The higher an object's speed and the greater its mass, the more kinetic energy it has.

Electrical energy
Electrical energy is the movement of electrons through a conductor. It is carried by an electric current to all kinds of appliances. Lightning occurs when electrons are discharged from a cloud.